

Overview & Scope

The Site Plan Agent acts as an orchestrator and generative AI solver for calculating buildable envelopes, optimizing architectural footprints, and routing site utilities. It directly encompasses two sub-agents:

- 1. Site Plan Agent (Core):** Ingests constraints and location data, uses shapely and spatial analysis to calculate a buildable WKT envelope, and runs an Adversarial AI loop (Generator/Corrector) to draft footprint coordinates that strictly adhere to constraints.
- 2. Tree Agent:** Evaluates existing trees on a lot, determines which can be preserved or removed based on the generated buildable envelope, calculates mitigation rules, generates a replanting topology, and creates a 2D rendering of the Tree Plan.
- 3. Civil Agent:** Analyzes public utility mainlines (water, sewer, stormwater), generates internal routing from the property boundary to the structure, constructs a 3D topographic terrain model from elevation point clouds, and generates both a Civil 2D CAD (.dxf) and an interactive 3D HTML viewer.

Core Terminology

To understand the limitations of the system, you must understand the distinction between these three geometric layers:

- **Property Boundary (Lot):** The absolute legal boundary lines of the parcel.
- **Buildable Envelope:** The invisible, legal polygon resulting from shrinking the Property Boundary inward by all required setbacks (front, rear, side). This can be any complex shape (e.g., triangle, weird polygon).
- **Exterior Shell (Footprint):** The actual physical walls of the proposed house generated by the AI, which must be placed 100% inside the Buildable Envelope.

Known Limitations & Edge Cases

1. Site Plan Agent: Geometric Math & Weird Shapes[Bottleneck - Weak LLM, could not make a more general approach work with the LLMs used, it was more reliable LLM(but weak) v/s not reliable (but better) LLM - I went with Weak but reliable(and fast), hence my architecture is built around this choice]

- **CRITICAL:** The A1Generator LLM heavily struggles to fit footprints into highly irregular or weirdly-shaped buildable envelopes (like triangles, cul-de-sacs, or diagonal boundaries).
- **Why?:** The LLM only receives the max bounding box width and depth of the envelope. It cannot natively visualize diagonal walls. Therefore, its generated footprint (even if requested to

be a simple rectangle) will frequently poke outside the diagonal lines of the legal envelope and get rejected endlessly by the Z3 Verifier.

2. Tree Agent: Grid Limits & Aggressive Removal

- Over-Removal: The agent decides to remove a tree if it intersects with the Buildable Envelope, rather than the actual Exterior Shell (Footprint). This guarantees safe construction, but it aggressively removes trees that the final house might never have touched.
- Replanting Grid: The replanting algorithm uses a strict 2ft coordinate grid. If a lot is extremely small or densely packed, it may run out of candidate points and fail to replant the required number of trees. All replanted trees are assumed to have a fixed DBH of 2.0.

3. Civil Agent: Straight-line Routing & Topography

- Stormwater Routing: The A* graph heavily penalizes uphill routes. If there is no natural downhill slope to a low point or city main, it might fail to route stormwater entirely (it does not currently fall back to proposing a pumped system).
- Sewer & Water Routing: These are routed using strict shortest-distance straight lines from the city mains to the building shell. They currently do not detect obstacles like preserved trees or steep cliffs along that straight line.

Bypassing Footprint Limitations (Weirdly-Shaped Lots)

If you are working on a highly irregular parcel (e.g., the Final Evaluation Project) and the AI Generator gets stuck in a rejection loop due to diagonal boundaries, there are three exact ways to bypass the issue:

Route 1: The Built-in HITL Route (For End-Users) **[NOT SUGGESTED FOR NOW]**

The agent pauses with a status of `waiting_for_user`. The user can send a `/api/v1/site-plan/resume` REST request passing explicit `user_feedback` containing exact custom coordinate points that bypass the AI's math struggles.

Route 2: The Hardcoded Override **[SUGGESTED]**

You can bypass the LLM entirely and manually inject the exact footprint coordinates you want for a specific lot.

- *File to Modify*: `src/agents/site_plan/agent.py`
- *Where*: Inside the `run_phase_3_ai_loop()` method, intercept the `current_code` returned by `a1.generate_topology()` or directly overwrite `shell_xy` with a hardcoded JSON array of points (e.g., `[[-10,-10], [10,-10], [10,10], [-10,10]]`).

Route 3: Modify the Generator Prompt [Will take time to tweak around with the prompts]

You can loosen the prompt restrictions to try and force the AI to mold to the exact envelope vertices rather than drawing rectangles.

- *File to Modify:* src/agents/site_plan/core/generator.py
- *Where:* Edit the shape_instructions and the PROMPT string in generate_topology(). Remove the rigid requirements for "rectangular", "L-shape", or "U-shape" and instruct the LLM to simply trace the provided envelope_coords scaled down by 10%.

Suggested:

Looking at the current timeline of final project completion you are highly suggested to do the Route 2 fix of this issue since most of the lots are irregular and it will be difficult for me or you to implement a prompt or A1/A2 architecture to account for the same and even then the limitation of LLM API currently used being weak and not strong enough to actually work, current system is designed to work with these weak LLMs.